Lecture 2.1 – First-Order Motor Response

Week 2: Connecting the Motor Model to Time Response

Learning Objectives

By the end of this lecture, you should be able to:

- explain what a step input is;
- describe the shape of a first-order motor-speed response;
- derive the first-order step-response equation;
- interpret the steady-state gain K;
- interpret the time constant τ;
- use the 63.2% rule to estimate τ from measured data; and
- estimate a simple motor transfer function from a step-response test.

1. Review: Our First-Order Motor Model

In Week 1, we introduced the simple first-order model

$$\boxed{G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}}$$

where

- $u(t)$ is the motor input, such as a PWM command;
- $y(t)$ is the motor output, such as speed in RPM;
- K is the steady-state gain;
- τ is the time constant.

The corresponding time-domain differential equation is

$$\boxed{\tau \frac{dy(t)}{dt} + y(t) = Ku(t).}$$

The transfer function tells us how the motor input and output are related. In Week 2, we focus on what that relationship looks like as a function of time.

2. What Is a Step Input?

A **step input** is a sudden change from one constant input value to another.
For example, the motor command may change from

$$u(t) = 0$$

to

$$u(t) = 160$$

PWM units.
A simple mathematical representation is

$$u(t) = U_0, \qquad t \geq 0,$$

where U_0 is the size of the step.
For a step of U_0,

$$U(s) = \frac{U_0}{s}.$$

Why do engineers like step tests?

- They are easy to apply.
- They reveal how quickly the system responds.
- They reveal the final steady-state output.
- They give us a practical way to estimate K and τ.

3. Deriving the First-Order Step Response

Start with

$$G(s) = \frac{K}{\tau s + 1}.$$

For a step input $U(s) = U_0/s$,

$$Y(s) = G(s)U(s) = \frac{K}{\tau s + 1}\frac{U_0}{s}.$$

Therefore,

$$Y(s) = \frac{KU_0}{s(\tau s + 1)}.$$

The inverse Laplace transform gives

$$\boxed{y(t) = KU_0\left(1 - e^{-t/\tau}\right), \qquad t \geq 0.}$$

This equation is the basic response we will use for the motor.

At $t = 0$,

$$y(0) = KU_0(1 - 1) = 0.$$

As $t \to \infty$,

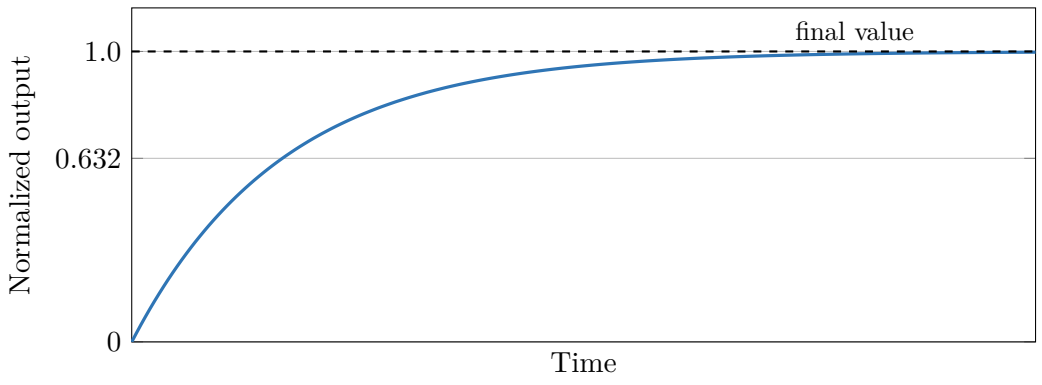
$$e^{-t/\tau} \to 0,$$

so

$$y(\infty) = KU_0.$$

Therefore the motor starts from its initial value and gradually approaches its final speed.

4. Shape of a First-Order Response



Important observations:

- The ideal first-order response does not jump instantly.
- The response changes fastest near the beginning.
- The response becomes slower as it approaches the final value.
- The ideal first-order response does not overshoot.

5. Steady-State Gain K

For a step input U_0,

$$y_{ss} = KU_0.$$

Therefore,

$$\boxed{K = \frac{y_{ss}}{U_0}.}$$

For our motor, a convenient unit is

$$\frac{\text{RPM}}{\text{PWM unit}}.$$

Worked Example 1

Suppose we apply

$$U_0 = 160$$

PWM units and measure a final motor speed of

$$y_{ss} = 125 \text{ RPM}.$$

Then

$$K = \frac{125}{160} = 0.78125.$$

Thus,

$$\boxed{K \approx 0.781 \ \frac{\text{RPM}}{\text{PWM unit}}.}$$

K describes "how much" output we obtain from the input. It does not describe how quickly the motor reaches that output.

6. Time Constant

The time constant describes how quickly the first-order system responds.
At

$$t = \tau,$$

the response is

$$y(\tau) = KU_0 \left(1 - e^{-1}\right).$$

Since

$$1 - e^{-1} \approx 0.632,$$

we obtain

$$\boxed{y(\tau) = 0.632\, y_{ss}.}$$

This gives us a very useful experimental rule:

Find the final speed. Calculate 63.2% of that speed. The time at which the response first reaches that value is approximately the time constant τ.

7. The 63.2 Percent Rule

Suppose

$$y_{ss} = 125 \text{ RPM}.$$

Then

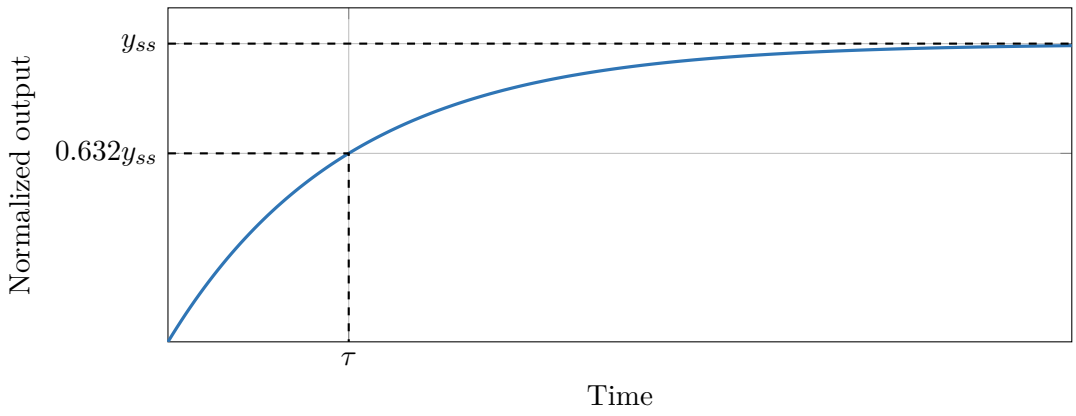
$$0.632 y_{ss} = 0.632(125) = 79.0 \text{ RPM}.$$

If the measured motor speed reaches approximately 79 RPM at

$$t = 0.42 \text{ s},$$

then

$$\boxed{\tau \approx 0.42 \text{ s}.}$$



8. Useful First-Order Benchmarks

For a first-order system,

Time	Fraction of final value reached
1τ	63.2%
2τ	86.5%
3τ	95.0%
4τ	98.2%
5τ	99.3%

A useful engineering approximation is:

$$\boxed{\text{first-order system is nearly settled after about } 4\tau.}$$

9. Estimating a Motor Transfer Function

A simple experimental procedure is:

1. Start the motor from rest.
2. Apply a known PWM step U_0.
3. Record motor speed versus time.
4. Estimate the final speed y_{ss}.
5. Calculate
$$K = \frac{y_{ss}}{U_0}.$$
6. Calculate
$$0.632 y_{ss}.$$
7. Find the time when the measured speed reaches that value.
8. Use that time as τ.
9. Form
$$\boxed{G(s) = \frac{K}{\tau s + 1}.}$$

10. Worked Example 2

A motor is tested with a step from PWM 0 to PWM 200.
The final measured speed is

$$y_{ss} = 155 \text{ RPM}.$$

First calculate the gain:

$$K = \frac{155}{200} = 0.775.$$

Now calculate 63.2% of the final value:

$$0.632(155) = 97.96 \text{ RPM}.$$

Suppose the speed reaches approximately 98 RPM at

$$t = 0.55 \text{ s}.$$

Therefore,

$$\tau \approx 0.55 \text{ s}.$$

The estimated motor model is

$$\boxed{G(s) = \frac{0.775}{0.55s + 1}.}$$

11. Python Connection

After estimating K and τ, Python can simulate the model.

```
import numpy as np
import matplotlib.pyplot as plt

K = 0.775      # RPM per PWM unit
tau = 0.55     # seconds
U0 = 200       # PWM step

t = np.linspace(0, 4, 400)
rpm = K * U0 * (1 - np.exp(-t / tau))

plt.plot(t, rpm, linewidth=2)
plt.grid(True)
plt.xlabel("Time (s)")
plt.ylabel("Motor Speed (RPM)")
plt.title("First-Order Motor Step Response")
plt.show()
```

The simulated final value should be approximately

$$KU_0 = (0.775)(200) = 155 \text{ RPM}.$$

12. Model Limitations

A real motor will not behave exactly like the ideal first-order model.
Possible differences include:

- friction and a minimum PWM needed to start rotating;
- encoder quantization;
- L298N voltage drop;

- changing supply voltage;
- motor heating;
- changing mechanical load.

A simple model is not meant to reproduce every detail. It should capture the main behavior well enough to help us understand and design the controller.

13. Check Your Understanding

A motor receives a step from PWM 0 to PWM 150.
The measured speed settles near 105 RPM, and the speed reaches 66.4 RPM after 0.40 s.
Find:

1. the gain K;
2. the time constant τ;
3. the transfer function $G(s)$;
4. the approximate settling time using 4τ.

Summary

- A step input is a sudden change in command.
- For a first-order motor model,
$$y(t) = KU_0(1 - e^{-t/\tau}).$$
- K determines the final output for a given input.
- τ determines how quickly the response changes.
- At $t = \tau$, the response reaches 63.2% of its final value.
- Step-response data can be used to estimate
$$G(s) = \frac{K}{\tau s + 1}.$$